

## **Introduction**

Transmission of infections in health care facilities can be prevented and controlled through the application of basic Infection Prevention and Control precautions which can be grouped into standard precautions, which must be applied to all patients at all times, regardless of diagnosis or infectious status, and additional (transmission-based) precautions which are specific to modes of transmission-airborne, droplet and contact

Standard precautions are the primary strategy for minimizing the risk of transmission of healthcare-associated infections. Standard precautions are work practices that provide a basic level of infection prevention and are applied to everyone, regardless of perceived or confirmed infectious status. The purpose is to break the chain of infection focusing particularly but not exclusively on the mode of transmission, portal of entry and susceptible host sections of the chain. This guideline applies to best practice amongst Healthcare Workers (HCWs) in caring for all patients within the hospital environment; in order to minimize the risk of transmission of infections between people within the hospital.

## **Purpose**

- To prevent the spread of cross infection.
- To uphold standards of safe practice.
- To protect both healthcare staff and every individual in their care at all times from the transmission of infection during care/hazardous procedures whether the risk is known or unknown.
- To ensure maximum protection of patients, visitors and staff without the need to disclose the individual's diagnosis or other information that may be confidential.
- To ensure evidence based research underpins best practice.

## **Roles and Responsibilities**

It is the responsibility of Department heads to ensure that healthcare workers who report to them adhere to this guideline. All healthcare workers who are involved in patient care have a responsibility to observe the precautions outlined in these guidelines at all times. Healthcare workers (HCWs) have a professional responsibility to ensure that protective clothing is worn appropriately.

## **Definitions**

Standard precautions consist of the following elements:

### **1. Hand hygiene, (refer to Hand Hygiene Policy);**

Hand Hygiene (HH) the most important procedure for preventing cross contamination. It is used to remove or kill microorganisms that colonize the hands of HCWs occurring directly from patient contact or indirectly via contact with the environment.

- Patients should be taught the importance of good hand hygiene. Visitors should be encouraged to use alcohol handrub or hand sanitizers when entering and leaving all health care settings.
- All cuts and abrasions should be covered with a waterproof dressing. Skin conditions should be reported to the Staff Health Clinic
- Methods of HH involve either antibacterial soap and water or alcohol-based waterless hand rub.
- Perform hand hygiene before and after patient contact, after contact with blood, body fluids, excretions, secretions, mucous membranes, broken skin or contaminated items and after gloves have been removed.
- Alcohol-based hand rubs/gels may be used as an alternative to hand washing if hands are visibly clean.
- Alcohol handrub& hand sanitizers should not be used when dealing with patients suffering from gastroenteritis,diarrheal disease or clostridium difficile
- Double gloving is recommended during some Exposure Prone Procedures (EPPs) or when attending major trauma incidents
- HCWs must follow the WHO's 5 moments for HH:
  1. Before patient contact
  2. Before clean/aseptic tasks
  3. After body fluid exposure risk
  4. After patient contact
  5. After contact with patient surroundings/environment

## **2. The use of personal protective equipment (PPE);**

PPE refers to a variety of Infection Prevention and Control barriers and respirators used alone, or in combination, to protect mucous membranes, skin, and clothing from contact with recognized and unrecognized sources of infectious agents in healthcare settings. It should be chosen according to the risk of exposure. Health care workers should assess whether they are at risk of exposure to blood, body fluids excretions or secretions and choose their items of personal protective equipment according to this risk.

- HCWs must have access to supplies of personal protective equipment (PPE) and materials.
- Designated containers (waste bins) for used disposable PPE should be placed in a location that is convenient to the site of removal to facilitate disposal and containment of contaminated materials.
- Hand hygiene is always the final step after removing and disposing of PPE.

Protective clothing which is provided for staff in areas where there is high risk of contamination (e.g. operating room) must be removed before leaving the designated area. Inappropriate wearing of PPE and specialist attire (e.g. wearing operating room attire in other wards, public areas of the hospital, or outside the facility) may contribute to a public perception of poor practice.

## **Gloves**

The use of gloves can reduce the risk of acquiring infection through direct skin contact between HCW and patients. Gloves should not be worn unnecessarily or as a substitute for hand decontamination as prolonged and irrational use may cause adverse reactions and skin sensitivity and risk of cross infection

- Gloves are a single use item
- Gloves can reduce the likelihood of contact dermatitis in staff exposed to chemical agent
- Gloves must be worn when direct contact with blood, body fluids, non-intact skin or mucus-membranes is anticipated
- Use non-sterile gloves for examinations and other clean procedures, and use sterile gloves for sterile procedures.
- Gloves must be changed between patients and different procedures on the same patient
- Gloves must not be worn when using computer keyboards, answering the phone, writing in patient's care records or serving meals
- Gloves must be disposed of in clinical waste bin
- Perform hand hygiene immediately on removal of gloves

Gloves should be discarded after each procedure/care activity. The re-use of gloves is not recommended for the following reasons:

- Glove integrity can be damaged if in contact with substances such as isopropanol, ethanol, oils and disinfectants
- Many gloves will develop micro-punctures very quickly and will no longer perform their barrier function
- There is a risk of transmission of infection
- Washing of gloved hands or using an alcohol handrub on gloves is considered unsafe practice.

## **Gowns/plastic aprons**

Disposable aprons are impermeable to bacteria and water and protect the areas of maximum potential contamination on the front of the body.

- Wear a gown/plastic apron to protect skin and clothing during procedures that may generate splashes or aerosolization of body substances and cause the soiling of clothes.
- Securely fasten the tabs/ties to keep the gown/plastic apron in place while performing patient care activities in the patient room/procedure area.
- Remove the gown/plastic apron by untying the tabs/ties and folding it away from you in an inside-out manner. Roll it into a ball and discard.
- Change the gown/plastic apron for each patient and/or procedure.
- Hand hygiene should be performed after removing the apron.
- Gloves/aprons are not to be worn after leaving the patient room or procedure area.

## **Masks**

Face masks and eye protection or a full face shield should be worn where there is a risk of blood, body fluids, secretions and excretions splashing into face and eyes.

**Masks** - are loose fitting, single-use items that cover the nose and mouth. They are not necessarily designed for filtration efficiency, or to seal tightly to the face.

**Respirators**(N95 mask) - are intended to help reduce the wearer's exposure to airborne particles. When worn correctly, they seal firmly to the face, thus reducing the risk of leakage. Mask should:

- Masks should be changed when they become soiled or wet
- Masks should never be reapplied after they have been removed
- Masks should not be left hanging around the neck
- Touching the front of the mask while wearing it should be avoided
- Hand hygiene should be performed upon touching or discarding a used mask.
- Surgical mask or respiratory mask (N-95 mask) are meant to be used as single use every after patient encounter
- Wear a respiratory mask (N95 mask) when indicated to enter an airborne isolation room, and remove it only when outside of the room.
- The fit of the respiratory masks (N95 mask) is critically important and every user should have been fit tested, trained in the use of the respirator (N95 mask) and record kept available. Additionally, a seal check should be carried out each time a respirator (N95 mask) is worn.

## **Protective eye/face wear**

- Masks may be used in combination with goggles to protect the mouth, nose and eyes, or a face shield may be used instead of a mask and goggles, to provide more complete protection for the face.
- The eye protection chosen for specific work situations (e.g., goggles or face shield) depends upon the circumstances of exposure, other PPE used, and personal vision needs.
- Personal eyeglasses and contact lenses are NOT considered adequate eye protection.
- Removal of a face shield, goggles and mask can be performed safely after gloves have been removed, and hand hygiene performed.
- The ties, ear pieces and/or headband used to secure the equipment to the head are considered “clean” and therefore safe to touch with bare hands.
- The front of a mask, goggles and face shield are considered contaminated
- Wash and disinfect visibly soiled reusable face shields or protective eyewear prior to reuse.

### **3. Aseptic non-touch technique (ANTT); (refer to Aseptic Technique Policy);**

Aseptic non touch technique is the method employed to maintain asepsis, protecting the patient from healthcare associated infections and staff from contamination from the patient's blood, body fluids and toxic substances. Aseptic non-touch technique is a standardised approach that staffs are taught to identify and protect the key parts of any procedure, perform effective hand hygiene, set up a non-touch technique and wear only the appropriate personal protective equipment

### **4. Waste management (refer to Hospital Waste Management Policy);**

All HCWs are responsible for the safe management and disposal of waste:

- All waste shall be segregated according to their category at the time and source of generation.
- Approved containers for the disposal of clinical and related waste are puncture resistant, leak proof and clearly labeled using color-coded and symbols as per waste management policy.
- Once full, containers are to be securely closed and transported to the designated waste storage area.
- Precautions will be used during all waste handling activities.
- Hand hygiene is to be performed after handling and disposing of all waste products.

### **5. The safe handling and disposal of sharps (refer to Hospital Waste Management Policy);**

All HCWs are required to ensure that risks from sharps injuries are adequately assessed and appropriate control measures are in place:

- Safety engineered devices are used where available
- No recapping of needles is to occur. If recapping is essential, then use a one-handed "scoop" technique.
- Sharps are not to be passed by hand. If necessary to pass a sharp, it must be via a designated sharps tray/kidney dish
- Do not place used sharp items on any environmental surface.
- The person generating the sharp is responsible for the safe disposal of sharps
- Sharps are disposed of in puncture-resistant containers as close to the point of use as possible.
- Close sharps containers when  $\frac{3}{4}$  full and remove for incineration.
- Occupational exposures/needle stick injuries must be reported to Staff Health Clinic and IP&C Department.

## **6. Reprocessing of reusable medical equipment and instruments; (refer to Reprocessing Policy)**

- Ensure medical devices labeled as “Single Use Only” are not reprocessed or reused.
- This Symbol means “Single Use Only” ②
- Remove organic material from critical and semi-critical instruments/devices using recommended cleaning agents before transfer to CSSD for high-level disinfection or sterilization.
- Ensure that reusable equipment is properly transported in leak-proof containers to CSSD for reprocessing before use with another patient.
- All clinical stock must be checked weekly to ensure that it is in date and rotated for use.

## **7. Medical Devices/Patient Care Equipment (refer to Low Level Disinfection of Medical Devices Policy);**

Reusable equipment can be a potential source of infection if not appropriately decontaminated after each use. Appropriate decontamination of healthcare equipment must be in line with Infection Prevention and Control principles and manufacturer’s instructions.

- Ensure “Reusable Equipment” is appropriately decontaminated between patients to reduce the potential risk of cross infection.
- The user of the device is responsible for ensuring that it is visibly clean and free from contamination with blood/body fluids following each procedure and prior to re-use
- The user must sign and date the appropriate tag/labels to confirm that cleaning has taken place. During decontamination, the user must check clinical equipment for signs of damage and send for repair or disposal if appropriate.

## **8. Respiratory hygiene and cough etiquette;**

Correct respiratory hygiene and cough etiquette is effective in decreasing the risk of transmission of pathogens contained in large respiratory droplets

- Cover nose and mouth with a tissue when coughing or sneezing, wiping and blowing the nose
- Dispose used tissue in the nearest waste receptacle.
- Clean hands with soap and water or antiseptic solution or with an alcohol-based hand rub after touching respiratory secretions or handling contaminated objects.
- Ask patients suffering from respiratory pathogens to wear a standard surgical face mask in the presence of others to prevent droplet transmission through coughing.

## **9. Patient placement (refer to Isolation Policy);**

Additional (transmission-based) precautions are taken while still ensuring standard precautions are maintained. There are three categories of Transmission-Based Precautions:

- Contact Precautions
- Droplet Precautions
- Airborne Precautions.

- Due to the lack of side rooms it is often necessary to prioritize which infectious patient most urgently required single room isolation. The prioritization of infectious patients for isolation depends on the route of infection (which determines infectiousness) and the extent of potential consequences of spread of infection (ability to infect staff as well as patients, severity of infection, availability of treatment or prophylaxis).
- Patients with infections that spread via airborne droplet nuclei are a major priority to isolate.
- When a single-patient room is not available, consultation with Infection Prevention & Control Department is recommended to assess the various risks associated with other patient placement options (e.g. cohorting patients with same infection).

#### **10. Routine environmental cleaning; (refer to Hospital Routine Environmental Policy);**

Infectious agents can be widely found in healthcare settings and there is a body of clinical evidence suggesting the association between poor environmental hygiene and the transmission of infectious agents in healthcare settings. Transmission of infectious agents from the environment to patients may occur through direct contact with contaminated equipment, or indirectly, for example, via hands that are in contact with contaminated equipment or the environment and then touch a patient.

Ensure all environmental surfaces, particularly those are high touched surfaces that contact with patients are routinely cleaned with hospital approved detergent and water and disinfected when required (Risk Stratification Matrix to Determine Frequency of Cleaning). For isolation rooms/bays/wards, refer to terminal cleaning procedure (SOP/IPC/01/Vers1.0)

#### **11. Appropriate handling of linen. (refer to Hospital Linen Management Policy);**

All HCWs must take reasonable steps to reduce the possible risk of cross infection to patients, other HCWs and laundry staff from used and infected linen. All linen should be appropriately segregated and bagged for transport to the laundry. When handling soiled, fouled and infected linen, disposable gloves and apron should be worn. Linen should be removed from the bed with care and placed immediately into the appropriate bag at the bedside and not on the floor or carried through the ward/department. Laundry staff must not manually open bags containing infectious linen

- Used, soiled and fouled linen:

Used linen should be placed in a white fabric laundry bag.

Soiled and fouled items should first be placed into a water soluble (red alginate) bag sealed/tied and then placed in a red fabric laundry bag.

- Infected linen:

Infected linen should be placed in a red water soluble (alginate) bag and should be placed in a red fabric laundry bag.

#### **12. Food and drinks at the work station**

Consumption of food and drinks in clinical areas with potential for exposure to blood or other infectious material or where the potential for contamination of work surfaces exist are prohibited. However, water bottles with protective lids, properly labeled with the employees name are allowed.

### **13. Laboratory specimens.**

All specimens are a potential infection risk therefore all specimens must:

- Wear gloves before obtaining laboratory specimens.
- Be collected in the correct container with the lid securely fastened
- Care should be taken not to contaminate the outer receptacle
- Remove gloves and perform hand hygiene once all laboratory specimens are in the appropriate containers
- Clearly labeled with the correct patient details
- Should be placed inside the specimen bag
- Been transported in a rigid container to the relevant laboratory or by pneumatic system

### **14. Mortuary Care**

Preparing the deceased for the morgue always involves the handling of blood, body fluids, and biological agents and may also involve exposure to life-threatening biological, chemicals, radiation, or electrical current. Use appropriate personal protective equipment (PPE) at all times. After the physician declares death, perform the following tasks to prevent exposure to blood and body fluid during transportation to protecting morgue personnel:

- Remove all disposable tubes and lines appropriately.
- Dress all wounds with impervious material to prevent oozing of body fluids or bleeding from wounds or previous catheter sites.
- Request an appropriately sized body bag and place the body in the bag.
- Follow the proper identification of the body, transportation, and documentation in the morgue.
- Identify patients with known infectious diseases and they should have body tags labeled with the appropriate category
- Inform the morgue supervisor if the deceased was known to harbor an infectious agent. (This information will also be confirmed in writing on the identification tag attached to the body bag.)
- Body parts (including placentas, stillborns, products of miscarriage, etc.) must be place in a red bag, labeled clearly, and stored in the refrigerator until delivery to the morgue.
- Remove disposable PPE and discard immediately after the task is completed.
- Place contaminated linen in a laundry bag and send to the laundry.
- Do not drink or eat inside the morgue.



## References

| Title of book/ journal/ articles/ Website                    | Author                                          | Year of publication | Page  |
|--------------------------------------------------------------|-------------------------------------------------|---------------------|-------|
| Standard Precaution Guideline                                | HSE West Mid-Western Regional Hospitals         | 2012                | 3-21  |
| Standard Precaution Policy                                   | Northamptonshire Healthcare                     | 2017                | 1-20  |
| Standard Infection Prevention and Control Precautions Policy | Doncaster and Bassetlaw Teaching Hospital       | 2018                | 1-17  |
| Standard Precautions Policy                                  | Benalla Health                                  | 2014                | 1-9   |
| Standard Precautions                                         | The GCC Infection Prevention and Control Manual | 2018                | 14-19 |

## Review / Revision History

| Revision History |                       |                      |                               |
|------------------|-----------------------|----------------------|-------------------------------|
| Revision         | Description of change | Reviewed by          | Revision Date                 |
| 1                | Initial release       | Dr. Zaina Al Maskari | 15 September 2019             |
| 2                | Second release        | Ahlam Al Bulushi     | 7 <sup>th</sup> December 2022 |
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